FSI-SLM: An End-to-End Digital Twin that Reads Fluid–Structure Interaction Experiments

A U-Net flow surrogate and a small language model that interprets the result — the whole pipeline runs on one laptop

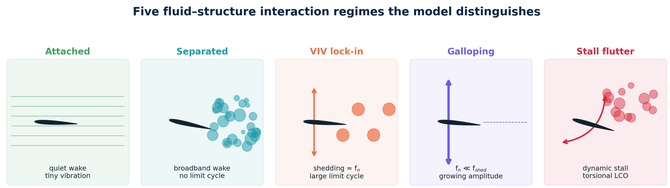
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1,928 solver cases • **107** airfoil sections • **R^2 = 0.998** lift, unseen shapes • **2.3%** velocity-field error • **F1 = 0.86** lock-in detection • **1** laptop, no GPU

MOTIVATION

Fluid–structure interaction decides when a hydrofoil, a riser or a turbine blade shakes itself apart. Reading what it produces — PIV fields, force traces, accelerometers — takes an expert, and the expert is the bottleneck. A frontier language model discusses the physics fluently, **invents** what it does not know, and needs a data centre to do it.



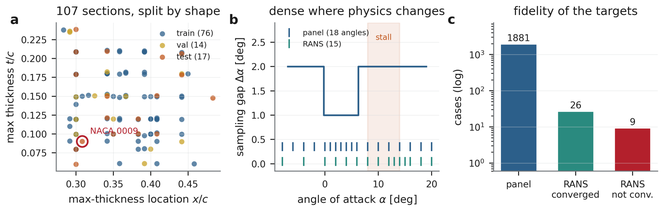
F1. The five flow–vibration regimes the interpreter must tell apart. Separating these by eye is the expert judgement we are automating.

OBJECTIVES

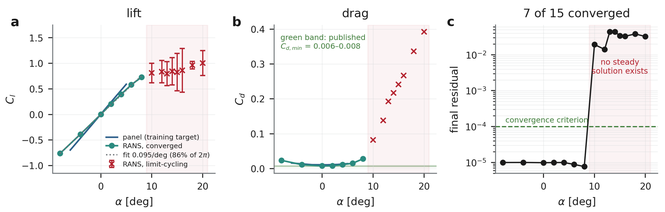
1. Reduce a raw FSI experiment to **one machine-readable record**.
2. Fine-tune a **small** language model to interpret the physics and flag likely sensor noise.
3. **Ground** every answer in published fluid dynamics by retrieval, so the model cites rather than invents.
4. Predict fields, forces **and** structural response straight from geometry, replacing the solver.
5. Serve all of it from a laptop over REST and MCP.

DATA

107 NACA sections at Re = 5×10^5, 18 angles of attack each. Three solvers write **one common tensor contract**, chosen per case by incidence: a Hess–Smith panel method with Beddoes–Leishman stall correction at any angle, steady RANS below stall, URANS above it where no steady solution exists. A chordwise Euler–Bernoulli beam turns surface pressure into load and displacement, so every case carries structure as well as flow.

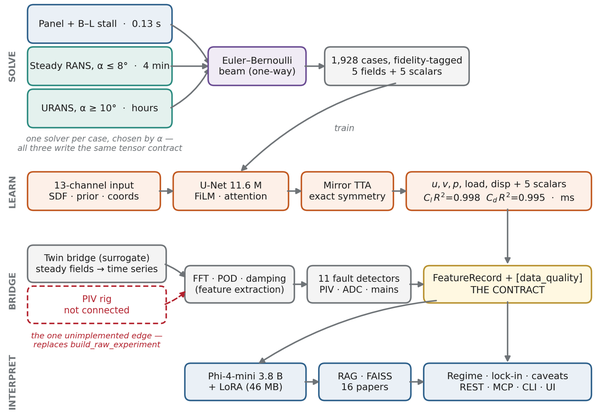


*F2. (a) The 107 sections by maximum thickness and its chordwise location. Splits hold out whole **shapes**, so NACA 0009 (circled) is pinned to test and every number reported for it is a generalisation result. (b) Angle sampling tightens from 2° to 1° through stall, where the physics changes fastest. (c) Target fidelity, log scale.*

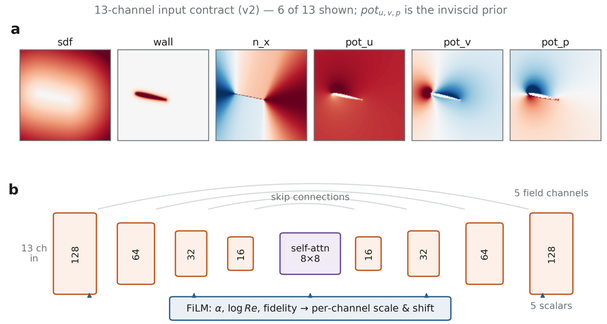


*F3. Are the targets right? Steady-RANS validation on the held-out section: 15 OpenFOAM cases, k-ω SST, 324 k cells. The converged branch gives a lift slope of 0.095/deg, 86% of thin-airfoil 2π, and a drag minimum of 0.0078 at α = 0, inside the published 0.006–0.008 band. Every case at α ≥ 10° hit the iteration cap: separated flow has no steady solution to converge **to. RANS validation is claimed for α ≤ 8° only**; the failed cases stay in the manifest with their residuals.*

METHODS



*F4. The twin end to end. The U-Net learns geometry → fields and coefficients; the twin bridge turns a steady field into a time-resolved record, and eleven fault detectors label PIV dropout, ADC clipping and mains hum before the model sees anything. A LoRA-tuned Phi-4-mini interprets it, grounded by retrieval over 16 papers. **The dashed red edge is the one unimplemented link:** a PIV rig would replace the synthetic experiment, and nothing downstream changes.*



*F5. (a) Six of the thirteen input channels. pot_u, pot_v, pot_p are the inviscid panel solution on the same raster, so the network predicts the **correction** to potential flow rather than the flow — stagnation, circulation and the far field arrive already exact. (b) 11.59 M parameters, self-attention at the 8×8 bottleneck because circulation is a global constraint; α, Re and fidelity enter by FiLM.*

ANALOGY: WHY SMALL + GROUNDED

A frontier model is a vast library with no lab notebook: fluent, unfalsifiable, data-centre-sized. Ours is a junior lab partner trained on **this** rig with the textbook open beside it — 46 MB of LoRA weights over a 3.8 B model and a FAISS index it must cite from. The U-Net is a learned wind tunnel: four minutes of RANS collapses to milliseconds, so a new case is a lookup, not an overnight solve.

REFERENCES

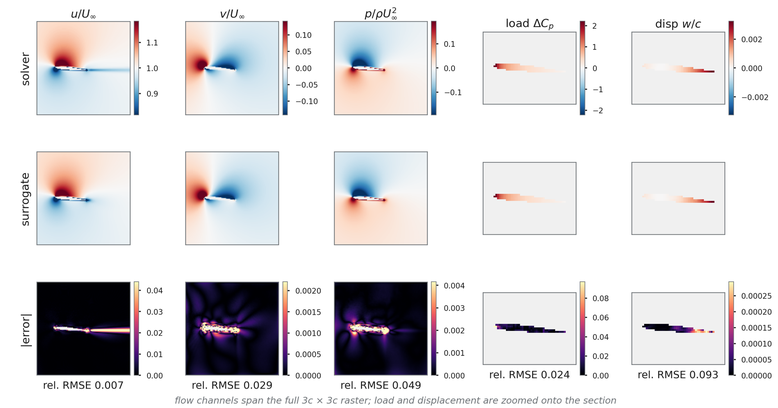
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ACKNOWLEDGEMENT

Thanks to Khemraj Shukla (Brown University) and the AGU Bright STaRS program. Built with Python, PyTorch, OpenFOAM, MLX and FAISS. Code and data: **github.com/AngadKumar16/FSI-SLM**

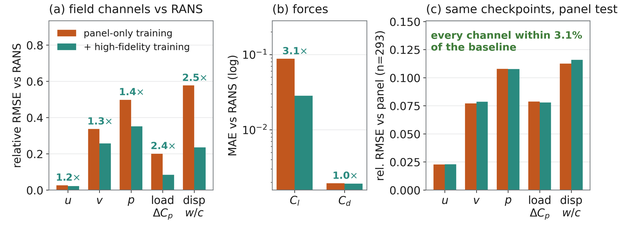
RESULTS

NACA 0009 (held out) at α = 5°, Re = 5×10^5

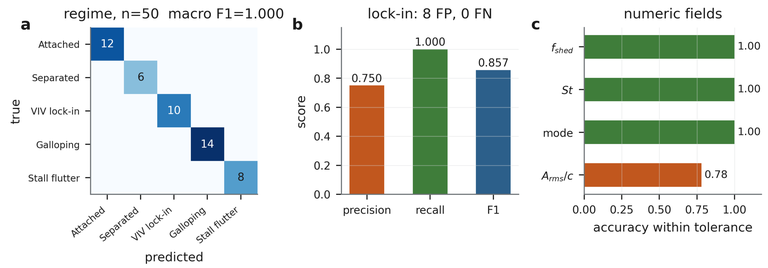


*F6. Solver, surrogate and error on NACA 0009 at α = 5° — a section held out of training entirely. Relative RMSE 0.007 (u), 0.029 (v), 0.049 (p), 0.024 (load), 0.093 (displacement); error concentrates in the near-wall layer and the wake. Across 293 held-out cases on 17 unseen sections: **R^2 = 0.998 for lift**, 0.995 for drag.*

Multi-fidelity fine-tuning: 47 high-fidelity cases (30 converged) on training-split sections close the transfer gap



*F7. Panel-only training reproduces RANS velocity and drag but not the structural channels — the chordwise beam amplifies the leading-edge pressure difference, exactly where the solvers disagree. Adding 47 high-fidelity cases on **training-split** sections closes most of the gap at no cost to panel accuracy — every field channel stays within 3.1% of the panel-trained baseline.*



*F8. The interpreter on 50 held-out records. The discriminating metric is lock-in detection: recall 1.000, precision 0.750 — it **never misses a lock-in** and over-calls eight. The rule-based labeller carries the same bias, so this was inherited, not invented.*

SUMMARY

The sensor half predicts velocity, pressure, load and displacement fields plus lift, drag and moment on shapes it has never seen. The interpreter half reads the record, names the regime, flags sensor faults and cites what it used. Both run on one laptop.

FUTURE WORK

Finish the URANS sweep so the deep-stall branch becomes trainable. Connect the PIV rig — the one dashed edge in F4. Close the structural loop so deflection feeds back into the flow. Add calibrated uncertainty on every field.